(1.a)

Given a vertex v, the program recursively runs on the children of the input vertex, with an input vertex with no children returning 1. This effectively counts all of the vertices, and each vertex is taken as an input to the function exactly once, the run-time of this function is just the number of vertices.

$\boxed{1.b}$

We use strong induction on the size of n

The smallest possible size of at least $2t+1 = 2(1)+1 = 3$

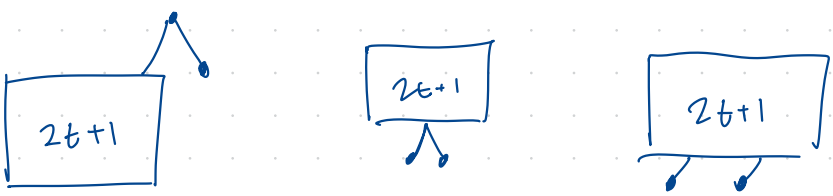
This is only possible tree of size 3

Each node has size 1

$t=1 \leq 1 \leq 2t-1 = 1$ ✓

Given that the prop. is true for a tree of size $n = 2t+1$,

Check $n = 2(t+1)+1 = 2t+1+2$



$t+1 < \text{v. size} < 2t+1$

Prove every binary tree of size at least $2t+1$ has vertex v s.t. $t \leq \text{v.size} \leq 2t-1$

Base case, let tree have size $2t+1$

Induction on t

Let $t=1$, $n=3$

End nodes have size at least 1

Assuming true for $t=s$, $t=s+1$

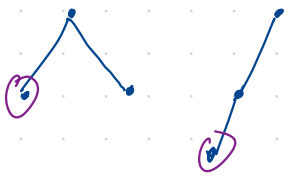
$2s+3$

Base case: let tree have size $n = 2t+1$

Now we have to prove proposition for $2t+1$

Let's do a second induction on t

$t=1$, $2t+1 = 3$

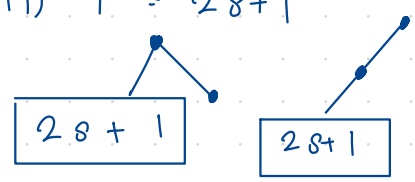


Encircled subtree has size $t=1 \leq 1 \leq 2t-1 = 1$

Now assume proposition true for $t = s$

Need subtree satisfying $s+1 < v.size < 2(s+1) - 1 = 2s+1$

Any tree of size $2(s+1)+1$ can be expressed as:



The boxed subtree of size $2s+1$ already satisfies

$s+1 \leq 2s+1 \leq 2s+1$

Hence it is true for $2t+1$, $t \geq 1$

Now going back to first induction

Assume proposition is true for $2t+1 \leq m \leq n$

then for a tree of size $n+1$

If the root has 0 children, then this case is not part of the induction hypothesis

If the root has 1 child, then that child is the root of a subtree of size n.

Then from the induction hypothesis this subtree has a vertex that satisfies the condition

If the root has 2 children, then checking cases:

one subtree has size at least $2t-1$: satisfied by induction hypothesis

one subtree has size $t \leq ____ < 2t-1$:

then $t \leq ____ < 2t+1 < n+1$

satisfied directly

one subtree has size $1 \leq ____ < t$

other subtree has size at least $2t+1-t = t+1$

this satisfies then the LHS of the condition

(not sure yet how to prove RHS)

2 (a) (i) $3(\log_2 n)^{3/2} = 3\left(\frac{\ln(n)}{\ln 2}\right)^{3/2} \longrightarrow (\ln(n))^{3/2}$

$g \rightarrow (\ln(n))^2$

$\frac{f}{g} \rightarrow \frac{\ln(n)^{3/2}}{\ln(n)^2} = (\ln(n))^{-1/2} = \frac{1}{\sqrt{\ln(n)}}$

The denominator is strictly increasing, $\therefore$ fraction $\rightarrow 0$

$f = o(g)$

$f = O(g)$

(ii) $f(n) = |\{$ 4 or less element subsets of n element set $\}|$

$= \cancel{\binom{n}{0}} + \binom{n}{1} + \binom{n}{2} + \binom{n}{3} + \binom{n}{4}$

leading term is $\binom{n}{4} \rightarrow \frac{n(n-1)(n-2)(n-3)}{4!}$

$\rightarrow$ leading term n^4

$n^4 > 3n^2$ for large n.

neither equality is satisfied

(iii) $\frac{n!}{2^{n^2}}$

Asked ChatGPT: "How to compare factorial with other functions?"
Wikipedia: Stirling's approximation

$\sim \frac{n^n}{2^{n^2}}$

taking nth root of top & bottom

$\rightarrow \frac{n^{\frac{n}{n}}}{2^{\frac{n^2}{n}}} = \frac{n}{2^n}$ Clearly the denominator grows much faster

$f = O(g)$

$g = o(g)$

(iv) Proof by counterexample:

Let $f = g$

$f = O(g)$, since $f \leq 2g$

but $\lim_{n \to \infty} \frac{f(n)}{g(n)} = 1$

$\therefore f \neq o(g)$, $g \neq o(f)$

I am particularly interested in the CS1200 topics on Intractability and Uncomputability.

Given my passion for quantum computing and how it has underpinned a significant portion of my coursework at Harvard, the question of intractability is a quintessential one. For example, one of the core questions discussed in Physics 160 was whether BQP, or the set of Bounded Quantum Probabilistic Polynomial Problems is strictly larger than the set of Bounded-error Probabilistic Polynomial time problems.

Thinking about uncomputability also interests me with the spreading influence of AI and the budding questions of AI safety and what it means to be human. I'd be interested in better circumscribing and delineating what are or will be limits of computation as we try to push those boundaries. In answering those questions, I think we can potentially understand better as well what it means to be human as technology becomes ever closer to our likeness.

As of now, I'm not apprehensive about much, but I was slightly concerned by this PSET 0 since I didn't have much time or support to work on it given that I arrived on campus late.